Fiscal Year:	FY 2019	**Task Last Updated:**	FY 03/19/2019
PI Name:	Mudawar, Issam Ph.D.		
Project Title:	Flow Boiling and Condensation Experiment (FBCE)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	47907-1288	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2017	**End Date:**	12/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Nahra, Henry K	**Contact Phone:**	216-433-5385
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hasan, Mohammad Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	NNX17AK98G		
Performance Goal No.:			
Performance Goal Text:			
	EDITOR'S NOTE: Continuation of "Flow Boiling and Condensation Experiment (FBCE)," grants NNX13AB01G and NNX12AK14G, which covered from 6/1/2012-5/31/2017. Several efforts are presently underway to reduce the size and weight of space systems by transitioning from single-phase to two-phase thermal management. Because of large density differences between liquid and vapor, buoyancy can play an important role in defining the motion of vapor relative to liquid and, therefore, influence heat transfer effectiveness. This is particularly the case for future space missions, including the manned mission to Mars. The emphasis in two-phase fluid physics is evident from several recent NASA workshops that culminated in critical recommendations concerning Rankine cycle power conversion, thermal control systems, and advanced life support systems. Among others, key fluid physics mechanisms that have been identified are flow boiling and condensation. A key strategy is to develop mechanistic models for flow conditions that would ensure insensitivity to gravity level for these important heat transfer		

Task Description:	mechanisms. This would allow performing most of the required confirmatory experiments on Earth. Those topics and the minimum flow requirements constitute key objectives for the proposed study. Unfortunately, most of the two-phase heat transfer know-how amassed over nearly a century of research comes from experiments conducted in Earth's gravity. The proposed study aims to develop an experimentally validated, mechanistic model for the minimum flow criteria to ensure gravity independent flow boiling heat transfer and condensation. Parabolic flight experiments and preparations for International Space Station (ISS) experiments will occur throughout the course of the proposed study. This project will be a joint effort between the Purdue University Boiling and Two-Phase Flow Laboratory (PU-BTPFL) and the NASA Glenn Research Center. Personnel from the two organizations combine extensive experience in research and development of both flow boiling and condensation systems, and in conducting microgravity experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research will lay the foundation of development of advanced high power to weight ratio thermal management systems for NASA's future space vehicles and planetary bases.
Task Progress:	This study explores subcooled nucleate flow boiling of FC-72 in a rectangular channel having two opposite heating walls at different mass velocities and wall heat fluxes ranging from 42% to 45% of critical heat flux (CHF). Experiments are performed in which constant and equal heat fluxes are supplied from the heating walls, with local wall temperatures measured at several axial locations and used to determine axial variations of local heat transfer coefficient. Additionally, detailed information on flow pattern and phase distribution is captured by high-speed video through the channel's transparent sidewalls. Experimental values for key fluid flow and thermal parameters are compared to predictions of 2D computational simulations using ANSYS FLUENT. The computations are based on the multi-phase volume of fluid (VOF) model combined with an appropriate phase change model, and account for conjugate heat transfer along the heating walls. Computed transverse profiles and stream-wise variations of void fraction, flow velocity, and wall temperature are examined. The computed results show reasonable agreement with experimentally captured interfacial behavior as well as heat transfer parameters. Computed cross-sectional temperature profiles for the mixture show non-equilibrium subcooled boiling with steep temperature gradients near the heating walls and low temperature liquid in the core region. This study is concluded by exploring limitations of the 2D model used.
Bibliography Type:	Description: (Last Updated: 01/06/2026)
Abstracts for Journals and Proceedings	O'Neill LE, Mudawar I, Hasan MM, Nahra HK, Balasubramaniam R, Mackey JR. "Experimental Investigation and Modeling of Density Wave Oscillations in Vertical Upflow Boiling." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Oct-2018
Abstracts for Journals and Proceedings	Nahra HK, Balasubramaniam R, Hasan MM, Mackey JR, Mudawar I, O'Neill LW, May RL. "Flow Boiling Experiments Using the Flow Boiling and Condensation Experiment (FBCE) Breadboard Test B." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Oct-2018
Abstracts for Journals and Proceedings	Mackey JR, Lant C, O'Neill LW, Mudawar I, Lokey A, Hasan MM, Nahra HK. "Test Section Image Quality Characterization of Flow Boiling Module for Microgravity Experiment." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Oct-2018
Articles in Peer-reviewed Journals	O'Neill LE, Mudawar I. "Mechanistic model to predict frequency and amplitude of Density Wave Oscillations in vertical upflow boiling." International Journal of Heat and Mass Transfer. 2018 Sep;123:143-71. Epub 2018 Mar 20. https://doi.org/10.1016/j.ijheatmasstransfer.2018.02.078 ; PubMed PMID: 30713350; PubMed Central PMCID: PMC6350787 , Sep-2018
Articles in Peer-reviewed Journals	Lee S, Devahdhanush VS, Mudawar I. "Investigation of subcooled and saturated boiling heat transfer mechanisms, instabilities, and transient flow regime maps for large length-to-diameter ratio micro-channel heat sinks." International Journal of Heat and Mass Transfer. 2018 Aug;123:172-91. https://doi.org/10.1016/j.ijheatmasstransfer.2018.02.020 , Aug-2018
Articles in Peer-reviewed Journals	O'Neill LE, Mudawar I, Hasan MM, Nahra HK, Balasubramaniam R, Mackey JR. "Experimental investigation of frequency and amplitude of density wave oscillations in vertical upflow boiling." International Journal of Heat and Mass Transfer. 2018 Oct;125:1240-63. Epub 2018 May 16. https://doi.org/10.1016/j.ijheatmasstransfer.2018.04.138 ; PubMed PMID: 30713351; PubMed Central PMCID: PMC6350798 , Oct-2018
Articles in Peer-reviewed Journals	O'Neill LE, Mudawar I, Hasan MM, Nahra HK, Balasubramaniam R, Mackey JR. "Flow condensation pressure oscillations at different orientations." International Journal of Heat and Mass Transfer. 2018 Dec;127:784-809. Epub 2018 Jul 30. https://doi.org/10.1016/j.ijheatmasstransfer.2018.07.072 ; PubMed PMID: 30705463; PubMed Central PMCID: PMC6349262 , Dec-2018
Articles in Peer-reviewed Journals	Lee S, Mudawar I. "Enhanced model for annular flow in micro-channel heat sinks, including effects of droplet entrainment/deposition and core turbulence." International Journal of Heat and Mass Transfer. 2019 Apr;133:510-30. https://doi.org/10.1016/j.ijheatmasstransfer.2018.12.074 , Apr-2019